

Experiment-5: Use Autopsy to Create a Case and Import Evidence

Course / Lab: Digital Forensics Lab

Experiment No.: 5

Title: Use Autopsy to Create a Case and Import Evidence

Aim

To analyze a forensic disk image using Autopsy and extract digital evidence.

Requirements

- Autopsy (installed on Windows, Linux, or macOS)
- Forensic disk image or data source to analyze

Description

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices. It provides modules for file system analysis, keyword search, timeline visualization, hash analysis, and reporting. This experiment demonstrates how to create a case, import evidence, analyze data, and generate reports using Autopsy.

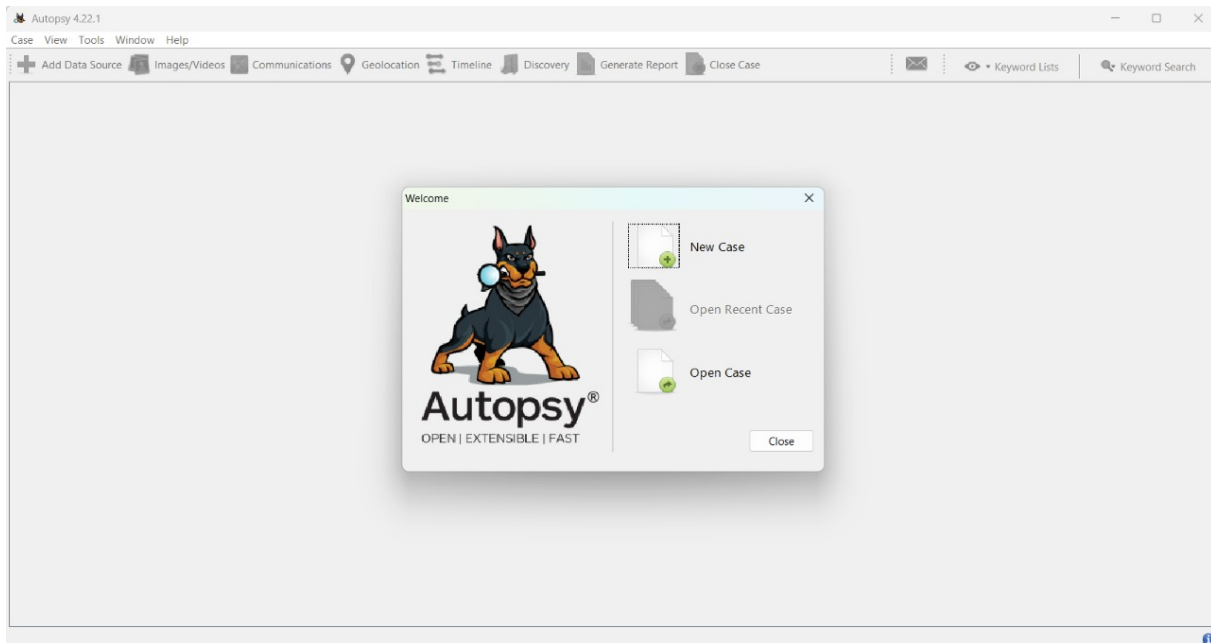
Procedure — Steps to Analyze Evidence Using Autopsy

Step-1: Installation

- Download and install Autopsy from the official website.
- Follow installation instructions based on your operating system (Windows, Linux, or macOS).

Step-2: Starting a New Case

- Create a new case by clicking on **New Case**.
- Enter the case name and location where case data will be stored.
- Fill in details such as case number, examiner's name, etc., and click **Next**.



Autopsy welcome screen – starting a new case

Step-3: Enter Case Details

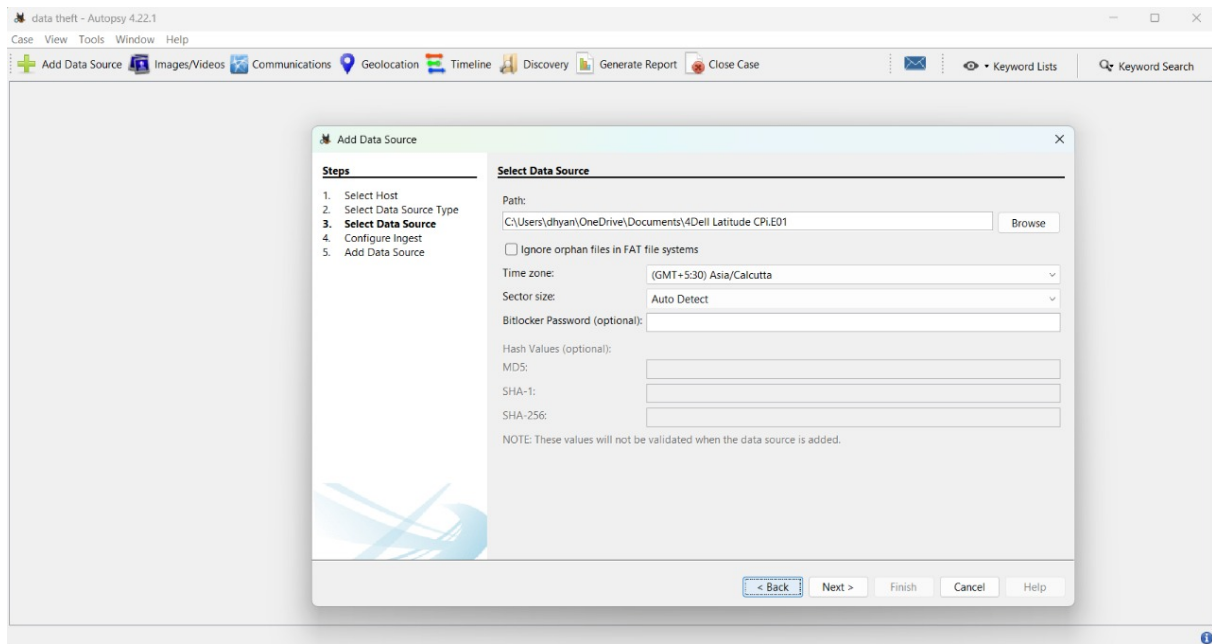
Enter the case name and location where the case data will be stored. Fill in the details like the case number, examiner's name, etc., and click Next.

The screenshot shows the 'New Case Information' dialog box. The title bar says 'New Case Information' with a close button. On the left, there is a 'Steps' section with two items: '1. Case Information' (selected) and '2. Optional Information'. The main area is titled 'Case Information' and contains the following fields and controls: 'Case Name:' with a text box containing 'data theft'; 'Base Directory:' with a text box containing 'C:\Users\dhyan\OneDrive\Documents\introduction' and a 'Browse' button; 'Case Type:' with two radio buttons, 'Single-User' (selected) and 'Multi-User'; and 'Case data will be stored in the following directory:' with a text box containing 'C:\Users\dhyan\OneDrive\Documents\introduction\data theft'. At the bottom of the dialog box are five buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

New Case Information dialog

Step-4: Adding a Data Source

- Choose the type of data source.
- Select the data source (e.g., disk image, local drive, logical files).
- Configure **Ingest Modules** (to analyze files, search keywords, extract metadata, etc.).
- Start the analysis.



Add Data Source – selecting the disk image and configuring options

Step-5: Initial Analysis and Overview

- Monitor **Ingest Progress**.
- Explore the resulting artifacts.
- Use the Tree Viewer to navigate evidence categories.

data theft - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

/img_4Dell Latitude CPLE01/vol_vol2

Table Thumbnail Summary

47 Results

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
OrphanFiles				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Allocated	Allocated
\$Extend				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	344	Allocated	Allocated
\$Unalloc				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Allocated	Allocated
[current folder]				2004-08-26 21:16:18 IST	2004-08-27 20:38:18 IST	2004-08-27 20:38:05 IST	2004-08-19 22:27:43 IST	168	Allocated	Allocated
Documents and Settings				2004-08-20 04:34:05 IST	2004-08-20 04:34:05 IST	2004-08-27 20:38:05 IST	2004-08-19 22:29:01 IST	56	Allocated	Allocated
My Documents				2004-08-20 20:51:05 IST	2004-08-20 20:51:05 IST	2004-08-20 20:51:09 IST	2004-08-18 22:25:24 IST	56	Allocated	Allocated
Program Files				2004-08-27 20:58:49 IST	2004-08-27 20:58:49 IST	2004-08-27 20:59:18 IST	2004-08-18 22:01:52 IST	152	Allocated	Allocated
RECYCLER				2004-08-25 21:48:25 IST	2004-08-25 21:48:25 IST	2004-08-27 20:42:30 IST	2004-08-25 21:48:25 IST	328	Allocated	Allocated
System Volume Information				2004-08-20 04:23:09 IST	2004-08-20 04:23:09 IST	2004-08-27 20:38:06 IST	2004-08-20 04:23:08 IST	160	Allocated	Allocated
Temp				2004-08-20 20:49:12 IST	2004-08-26 20:35:22 IST	2004-08-20 20:56:37 IST	2004-08-20 20:49:12 IST	48	Allocated	Allocated
WIN98				2004-08-18 21:58:38 IST	2004-08-18 21:58:38 IST	2004-08-20 20:56:37 IST	2004-08-18 21:58:38 IST	56	Allocated	Allocated
WINDOWS				2004-08-25 21:26:09 IST	2004-08-25 21:26:09 IST	2004-08-27 20:38:06 IST	2004-08-19 22:20:37 IST	56	Allocated	Allocated
\$AttrDef				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2560	Allocated	Allocated
\$BadClus				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	0	Allocated	Allocated
\$BadClus\$Bad				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	4869332992	Allocated	Allocated
\$Bitmap				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	1188808	Allocated	Allocated
\$Boot				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	8192	Allocated	Allocated
\$LogFile				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	26460160	Allocated	Allocated
\$MFT				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	12600320	Allocated	Allocated
\$MFTMirr				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	4096	Allocated	Allocated
\$Secure\$SDS				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	277616	Allocated	Allocated

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Page: 1 of Page: Go to Page: 1 Jump to Offset: Launch in HxD

0000000001 33 C0 8E D0 8C 09 7C FB 50 07 50 1F FC 8E 1B 7C 3.....P.P....

Analyzing files from 4Dell Latitude CPLE01 37% 6

File listing view with ingest in progress

Step-6: Detailed Analysis

Keyword Search:

- Perform searches using the Keyword Search module.
- Use pre-configured lists or enter custom keywords.

File Analysis:

- Navigate files and folders under File Types or File System.
- Open, view, or export files for further examination.

Timeline Analysis:

- Use the Timeline module to visualize events by timestamps.
- Helps track user activity over time.

Hash Analysis:

- Compare file hashes against known databases to identify known good/bad files.

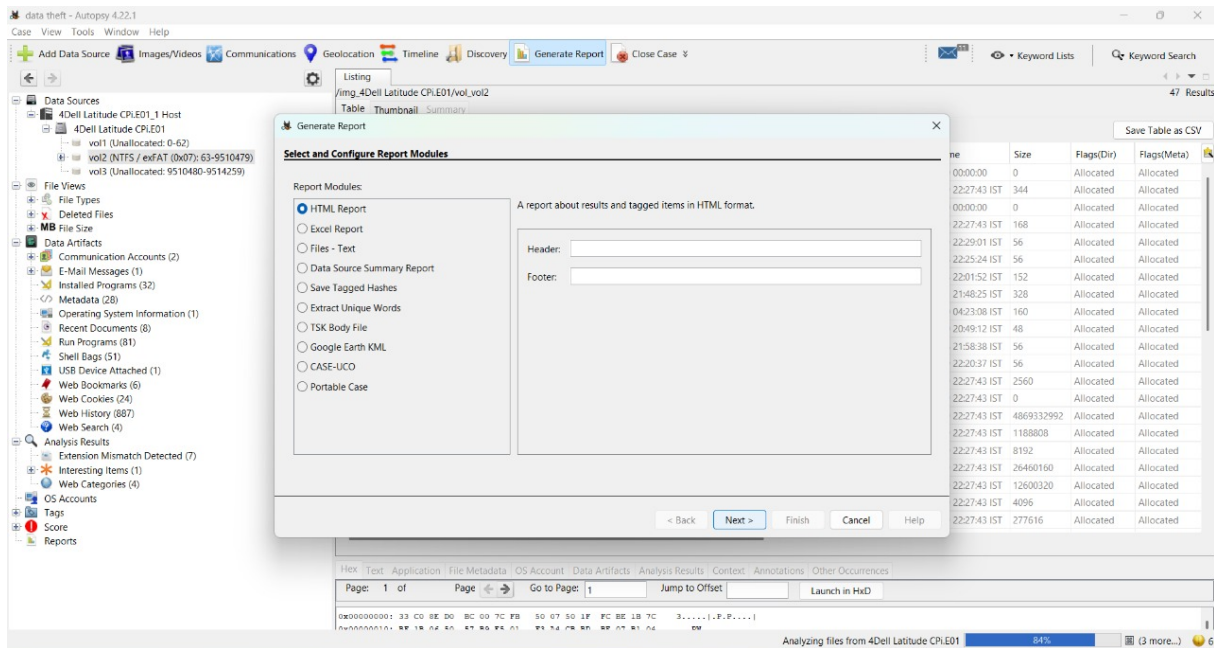
The screenshot displays the Autopsy 4.22.1 interface. The left sidebar shows the 'Data Sources' tree with '4Dell Latitude CPLE01' selected. The main window is titled 'Listing' and shows a table of files from '4Dell Latitude CPLE01/vol2'. The table has columns for Name, S, C, O, Modified Time, Change Time, Access Time, and Created. The files listed include system files like \$orphanFiles, \$Extend, \$Unalloc, and various user files like Documents and Settings, My Documents, Program Files, RECYCLER, System Volume Information, Temp, WIN98, WINDOWS, \$AttrDef, \$BadClus, \$Bitmap, \$boot, \$LogFile, \$MFT, \$MFTMirr, and \$Secure\$SDS. The bottom status bar indicates 'Analyzing files from 4Dell Latitude CPLE01' with a progress bar at 67% and a button to 'Launch in HxD'.

Name	S	C	O	Modified Time	Change Time	Access Time	Created
\$orphanFiles				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
\$Extend				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST
\$Unalloc				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
[current folder]				2004-08-26 21:16:18 IST	2004-08-27 20:38:18 IST	2004-08-27 20:38:05 IST	2004-08-27 20:38:05 IST
Documents and Settings				2004-08-20 04:34:05 IST	2004-08-20 04:34:05 IST	2004-08-27 20:38:05 IST	2004-08-27 20:38:05 IST
My Documents				2004-08-20 20:51:05 IST	2004-08-20 20:51:05 IST	2004-08-20 20:51:09 IST	2004-08-20 20:51:09 IST
Program Files				2004-08-27 20:58:49 IST	2004-08-27 20:58:49 IST	2004-08-27 20:59:18 IST	2004-08-18 22:01:52 IST
RECYCLER				2004-08-25 21:48:25 IST	2004-08-25 21:48:25 IST	2004-08-27 20:42:30 IST	2004-08-25 21:48:25 IST
System Volume Information				2004-08-20 04:23:09 IST	2004-08-20 04:23:09 IST	2004-08-27 20:38:06 IST	2004-08-20 04:23:08 IST
Temp				2004-08-20 20:49:12 IST	2004-08-26 20:35:22 IST	2004-08-20 20:56:37 IST	2004-08-20 20:49:12 IST
WIN98				2004-08-18 21:58:38 IST	2004-08-18 21:58:38 IST	2004-08-20 20:56:37 IST	2004-08-18 21:58:38 IST
WINDOWS				2004-08-25 21:26:09 IST	2004-08-25 21:26:09 IST	2004-08-27 20:38:06 IST	2004-08-19 22:20:37 IST
\$AttrDef				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST
\$BadClus				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST
\$BadClus\$Bad				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST
\$Bitmap				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST
\$boot				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST
\$LogFile				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST
\$MFT				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST
\$MFTMirr				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST
\$Secure\$SDS				2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST	2004-08-19 22:27:43 IST

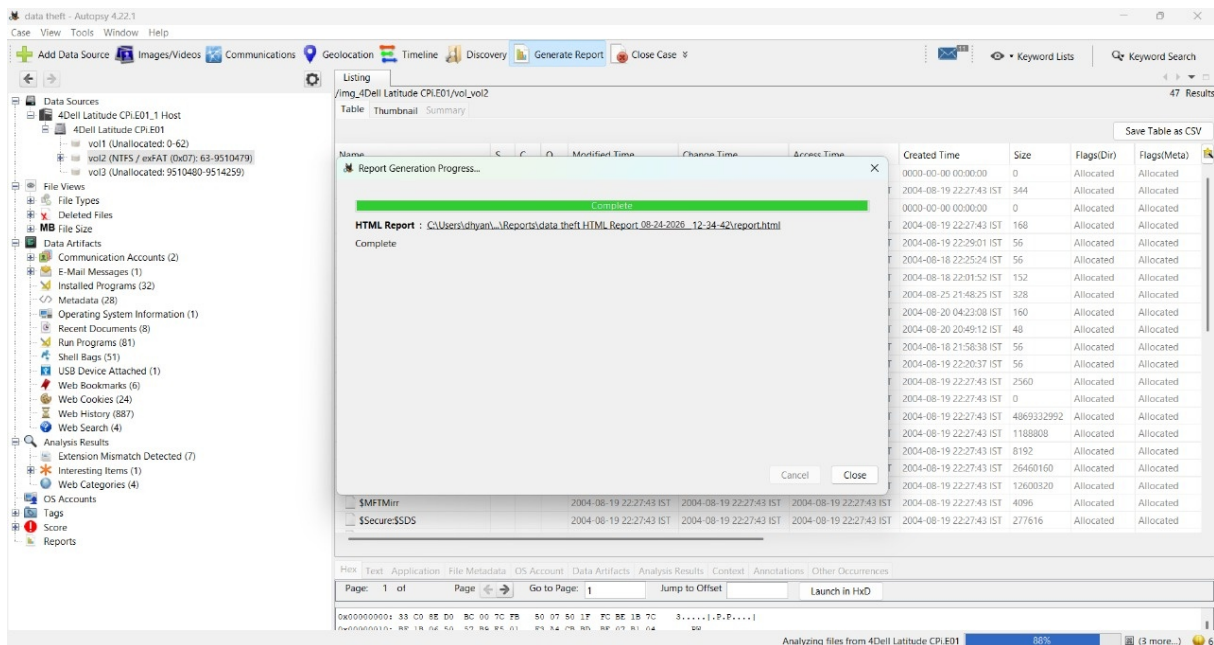
Keyword Search module in use

Step-7: Reporting

- Generate a report by selecting **Generate Report** from the toolbar.
- Choose report type (HTML, CSV, Excel, etc.).
- Select which analysis results to include.
- Export findings: export individual files or artifacts for reporting or further analysis.
- Final Review: check that the report includes all relevant details.
- Save or print the report for case documentation.

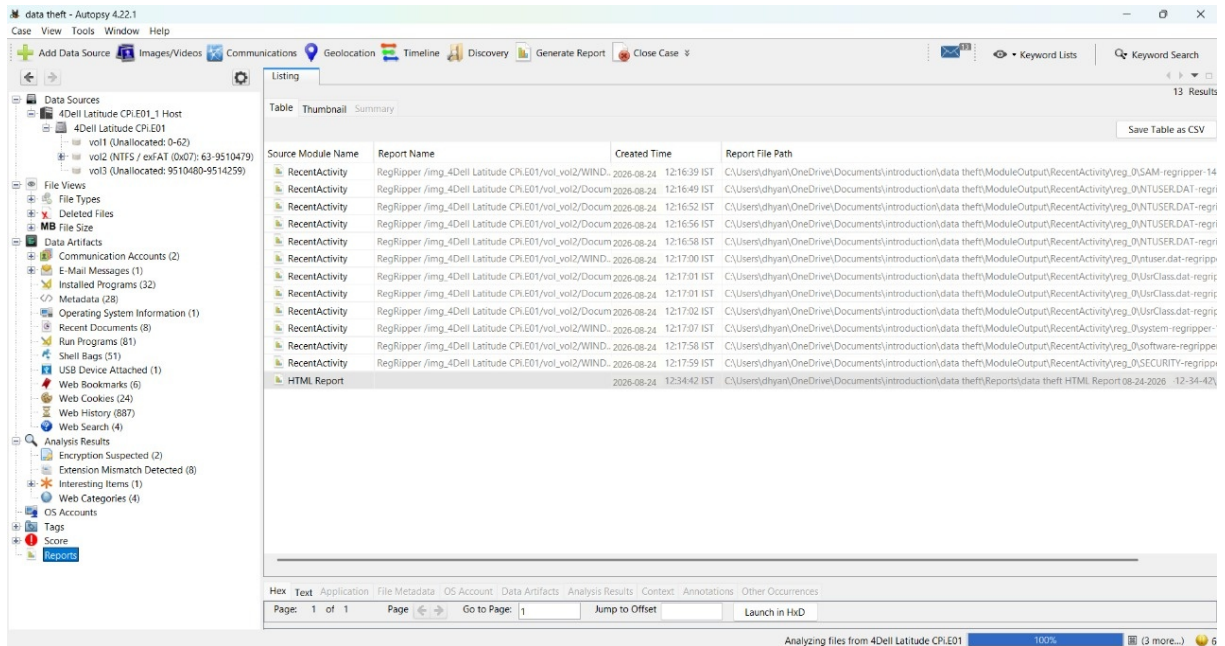


Generate Report – selecting the HTML Report module



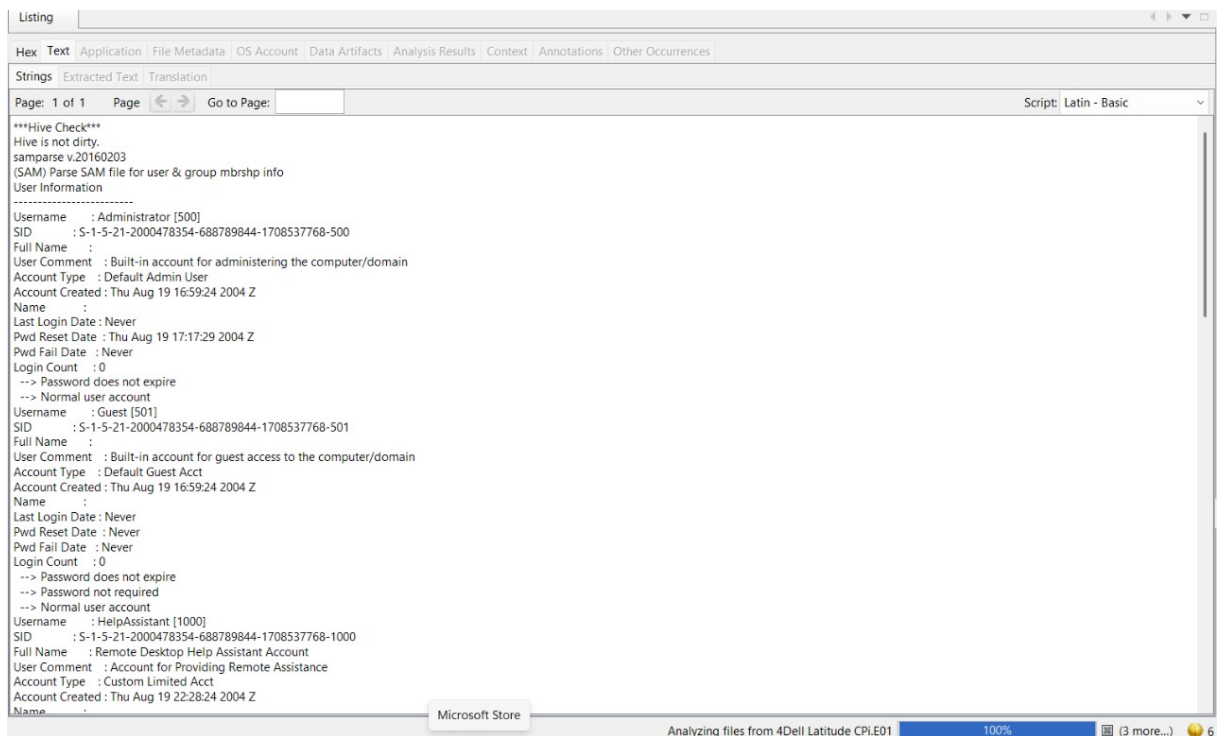
Report Generation Progress – completed successfully

Reports



Generated reports listing, with RegRipper and HTML report entries

Text in the report



Extracted SAM registry hive content from the generated report

Step-7: Case Closure

- Close the case once the investigation is complete.
- Archive all data and reports according to organizational policies.

Result

The experiment successfully demonstrated the use of Autopsy for creating a case, importing evidence, analyzing digital artifacts, and generating reports, thereby validating its effectiveness as a digital forensic investigation tool.